

# Technical Document #367: Cloud Computing - Comprehensive Overview

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Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

### Key Takeaways:

- Kubernetes orchestrates containerized applications across clusters of machines.
- Auto-scaling automatically adjusts compute resources based on demand.
- Containerization packages applications with their dependencies.